

SAHIL BHOIR

Software Engineer | Applied Machine Learning | Backend and AWS

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SUMMARY

I am a Computer Engineering graduate focused on applied machine learning, backend engineering and AWS. I build complete systems around models - from reliable data and realistic evaluation to APIs, cloud workflows, testing and user-facing interfaces. My work combines measurable results with clear operational limits, including performance on unfamiliar data and uncertainty from pretrained services.

PROJECTS

Cyberattack Tactics Triage - [Python](#) · [TF-IDF](#) · [Linear SVM](#) · [Calibration](#) · [MITRE ATT&CK](#) | [GitHub](#)

- Achieved 0.773 macro F1, 0.826 micro F1 and 0.720 exact match across 15 ATT&CK tactics by training a multilabel Linear SVM on 16,955 cleaned Enterprise ATT&CK procedure examples.
- Strengthened evaluation reliability by replacing a random split with source-grouped partitions, removing eight cross-partition near-duplicates and testing on 3,391 grouped examples.
- Built an analyst-review workflow with calibrated confidence, feature-level explanations and CSV batch processing; an unseen-technique stress test fell to 0.516 macro F1, making the boundary on unfamiliar behaviour explicit.

Real-Time Violence Event Detection - [YOLOv8](#) · [OpenCV](#) · [FastAPI](#) · [Streamlit](#) | [Demo](#) · [GitHub](#)

- Selected a stable event-filter configuration from 17,289 frame observations across 35 reviewed clips using a 0.70 confidence threshold, five positive activation frames and three negative release frames.
- Designed an N-positive/K-negative state machine that converted noisy pretrained YOLOv8 frame predictions into persistent, reviewable events while suppressing duplicate alerts.
- Delivered an end-to-end monitoring workflow for video, webcam and RTSP inputs with FastAPI controls, Streamlit review, stored screenshots, runtime diagnostics and optional Telegram alerts; detector-level misses remained separate from temporal-filter performance.

Serverless Receipt Processor - [AWS Lambda](#) · [Cognito](#) · [S3](#) · [Step Functions](#) · [Textract](#) · [DynamoDB](#) | [Demo](#) · [GitHub](#)

- Delivered an authenticated receipt-to-expense workflow that transformed uploaded images into user-scoped, editable records using Cognito, S3, Textract, Lambda and DynamoDB.
- Separated upload and processing workloads through short-lived presigned URLs, EventBridge and Step Functions, keeping asynchronous OCR and parsing outside the client request path.
- Improved extraction reliability with five focused parser regressions for ambiguous totals and Indian currency and date formats, while retaining the original receipt and editable fields when OCR remained uncertain.

EDUCATION

Vidyalankar Institute of Technology - B.E. Computer Engineering, AI and ML Minor | **CGPA: 9.43/10** | 2021-2024

TECHNICAL SKILLS

Programming: Python, JavaScript, Java, C/C++, SQL

Web and Backend: HTML, CSS, FastAPI, Flask, REST APIs, WebSockets

Cloud and Data: AWS Lambda, S3, DynamoDB, API Gateway, Step Functions, Pandas, NumPy

ML and Engineering: PyTorch, scikit-learn, XGBoost, Transformers, OpenCV, Docker, Git/GitHub, Linux, automated testing, Data Structures and Algorithms, OOP

CERTIFICATIONS

Certifications: AWS Certified Cloud Practitioner (CLF-C02) · IBM Full Stack Software Developer Professional Certificate

ADDITIONAL

Other Work: [Portfolio](#)

Writing: [Medium](#)